Physics-Guided Neural Network for Correcting Residual Ionospheric Error in GNSS Radio Occultation Bending Angle

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BIOGRAPHY

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ABSTRACT

Global Navigation Satellite System (GNSS) radio occultation (RO) achieves high accuracy retrievals in the upper troposphere and low stratosphere. However, residual ionospheric errors (RIE), which are remaining errors after the standard ionospheric correction of dual-frequency bending angles, emerge as systematic biases limiting retrieval accuracy toward higher altitudes (above ~35 km). The ray path separation of dual-frequency signals is the main contributor to RIE. Kappa correction, theoretically derived under the assumption of spherical symmetry to correct for such effects, exhibits reduced performance in mitigating RIEs in the presence of ionospheric horizontal gradients. To address this limitation, we propose a physics-guided deep learning framework named PhyG-TRF (Physics-Guided Transformer) that refines RIE corrections by integrating the physics baseline (kappa correction) with an altitudinal sequence learning approach using a Transformer-based residual architecture. By treating each occultation event as a vertical sequence, PhyG-TRF captures both altitudinal dependencies and global environmental context—such as solar zenith angle and geomagnetic latitude—through a Multi-Head Self-Attention (MHSA) mechanism. A case-level adaptive weighting scheme based on the empirical CDF of the positive gradient ratio further enforces robustness in ionospheric asymmetry cases. Additionally, a Vector Adaptive Residual Layer (VARL) combines the physics baseline and the learned residual via a learned scaling gate, thereby capturing the complex asymmetry of the ionosphere. Utilizing simulated datasets generated via ray-tracing on a 3D ionospheric model, our framework demonstrates improved performance over the physics baseline and other machine learning models (XGBoost, Bi-GRU). Statistical analysis confirms that PhyG-TRF achieves a 52.1% reduction in RMSE over the physics baseline on the full test set (coefficient of determination (R^2) improving from 0.254 to 0.828). For the positive-trending RIE subset, RMSE is reduced by 70.0% and R^2 improves from −2.52 to 0.684. The altitudinal sequence learning strategy further provides superior structural smoothness over point-wise models, as demonstrated by jitter analysis. These results highlight the model's robust capability to resolve RIE, even in asymmetric ionospheric environments.

1 INTRODUCTION

Global Navigation Satellite System (GNSS) radio occultation (RO) has been proven useful for applications such as numerical weather prediction and climate monitoring. This technique utilizes the refraction and delay of GNSS signals to retrieve profiles of temperature and pressure in the neutral atmosphere. To ensure accurate retrievals, ionospheric effects must be removed, typically by applying a linear combination of dual-frequency (e.g., GPS frequencies of L1 (1575.42 MHz) and L2 (1227.60 MHz)) bending angles. The standard dual-frequency correction (Vorob'ev and Krasil'nikova, 1994) effectively eliminates the first-order ionospheric effect. However, because L1 and L2 rays travel different paths due to the dispersive nature of the ionosphere, a systematic bending angle bias remains even after correction. This residual, referred to as the residual ionospheric error (RIE), degrades the accuracy of retrieved atmospheric profiles particularly in the 40-80 km altitude region. We note that in this paper, RIE refers specifically to the ray path separation effect, neglecting errors associated with higher-order terms in the ionospheric refractive index.

Extensive theoretical considerations and empirical analyses have been conducted to characterize and mitigate the RIE (Danzer et al., 2013, 2020, 2021; Healy & Culverwell, 2015; Li et al., 2020). Notably, Healy and Culverwell (2015) introduced the "kappa correction" to account for RIE which depends on the squared electron density, theoretically derived under a spherical symmetry assumption. Subsequent studies have proposed refining the scaling coefficient (kappa) in the kappa correction as a function of various ionospheric parameters. Angling et al. (2018) proposed a linear approximation of kappa as a function of the F10.7 solar flux index, solar zenith angle (SZA), and impact parameter. Park et al. (2025) extended this study by incorporating geomagnetic latitude and local time into a four-condition (day/night × high/low solar activity) regression model. While these empirical models are easy to apply and improve upon the physics baseline, they remain limited in their ability to represent highly nonlinear RIE behaviors under strong ionospheric asymmetry.

Simulation studies indicate that RIE trends in complex, asymmetrical ionospheric conditions diverge from the patterns observed under spherical symmetry (Li et al., 2020; Liu et al., 2020; Wu et al., 2025; Syndergaard and Kirchengast, 2022). Unlike the spherically symmetric case where RIE typically manifests as a negative bias increasing in magnitude with altitude, these studies demonstrate that asymmetrical conditions—such as those induced by the Equatorial Ionization Anomaly (EIA) or day-night terminators—can induce different error characteristics, including positive biases in the presence of strong horizontal gradients. In such cases, the existing kappa correction—designed to subtract a negative bias—can inadvertently amplify the error rather than reduce it. This limitation is particularly critical during solar maximum periods, where high solar activity and severe ionospheric gradients are expected to produce complex RIEs that exceed the correction capability of simple physics models.

Recent research has increasingly applied deep learning to solve nonlinear modeling challenges specifically in GNSS-RO data processing. Lasota (2021) and Li et al. (2025) have demonstrated the potential of neural networks in GNSS-RO for tropospheric profiling and nonlinear bias removal. Aligning with this trend, this study employs deep learning to explicitly model RIE. The ability of neural networks to model highly complex patterns makes them a promising alternative to simplified physical models. Nevertheless, applying a simple "black box" model presents a challenge as it inherently fails to leverage the valuable baseline performance already provided by established physical models. Furthermore, purely data-driven approaches often struggle with the tail risk of error distributions, where rare but critical high magnitude errors are under-represented in training data.

Building on the concept of Physics-Guided Neural Networks (PGNN) (Daw et al., 2021), we propose PhyG-TRF (Physics-Guided Transformer), a robust RIE correction framework that integrates physical knowledge with deep sequence learning using Transformer-based architecture. By treating the vertical profile of a single occultation event as a sequence of altitude steps, PhyG-TRF leverages self-attention mechanism to aggregate global context from the entire altitude profile simultaneously, capturing complex altitudinal dependencies that point-wise or local sequential models cannot represent. Adopting the Hybrid-Physics-Data Residual (HPD-Res) (Daw et al., 2021) strategy, the model augments its input with the physics-based kappa correction estimate while learning to predict the residual error. A Vector Adaptive Residual Layer (VARL) then combines the physics baseline and learned residual through a learned gating mechanism, enabling the model to optimally weight physical knowledge against data-driven correction. This work is trained and validated using a 3D ray-tracing simulation RIE dataset generated for the year 2024, selected to include high solar activity conditions of Solar Cycle 25.

The outline of this paper is as follows. Section 2 details the methodology, covering the theoretical limitations of existing models, the dataset construction, the input feature design, and the PhyG-TRF architecture. Section 3 outlines the experimental setup, including the definition of baseline models, data partitioning strategies, and training configurations. Section 4 presents a comprehensive evaluation of PhyG-TRF. Finally, conclusions and discussions are provided in Section 5.

2 METHODOLOGY OF THE RESIDUAL IONOSPHERIC ERROR CORRECTION

In this study, we present a physics-guided deep learning framework, named PhyG-TRF, designed to overcome the limitations of traditional ionospheric correction methods. Before detailing the dataset and model architecture, we first review the physical definition of the RIE and existing correction models to establish the physics baseline used in our framework.

2.1 Theoretical Background and Problem Formulation

The standard approach for removing the ionospheric contribution from GNSS-RO measurements involves taking a linear combination of the bending angles measured at two frequencies (e.g., L1 and L2), interpolated to a common impact parameter. As proposed by Vorob'ev and Krasil'nikova (1994), the ionosphere-corrected bending angle, $\alpha_{LC}(a)$, is given by:

$$\alpha_{LC}(a) = \frac{f_1^2\alpha_1(a) - f_2^2\alpha_2(a)}{f_1^2 - f_2^2} = \alpha_{NA}(a) + \alpha_{RIE}(a) \tag{1}$$

where f_1 and f_2 denote the signal frequencies, and $\alpha_1(a)$ and $\alpha_2(a)$ are the observed bending angles at impact parameter a. Here, $\alpha_{NA}(a)$ represents the bending angle induced solely by the neutral atmosphere, which is the primary target quantity required for accurate retrieval of atmospheric parameters such as temperature and pressure. Ideally, the first-order correction in Eq. (1) would isolate $\alpha_{NA}(a)$ by removing the dominant ionospheric effect proportional to $1/f^2$. However, a residual error denoted as $\alpha_{RIE}(a)$ remains due to the ray separation between the dual-frequency signals. Other higher-order contributions, such as geomagnetic terms, are neglected in this paper.

To mitigate the RIE, Healy and Culverwell (2015) proposed a modification to the standard correction, introducing a term dependent on the squared difference of the L1 and L2 bending angles. This method, known as kappa correction, is expressed as:

$$\alpha_{corr}(a) = \alpha_{LC}(a) + \kappa(a) \cdot [\alpha_1(a) - \alpha_2(a)]^2 \tag{2}$$

where $\kappa(a)$ is a scaling coefficient. Analytically derived under the assumption of a spherically symmetric ionosphere modeled by a single Chapman layer, $\kappa(a)$ is a function of the signal frequencies, Chapman layer peak height (r_m), impact parameter (a), and the ionospheric scale height (H):

$$\kappa(a) = \left(\frac{f_1^2 f_2^2}{f_1^2 - f_2^2}\right)^2 \left(\frac{1}{2\pi H}\right)\left(\frac{\sqrt{r_m^2 - a^2}(2r_m^2 + a^2)}{4ar_m}\right) \tag{3}$$

While Eq. (3) provides a physics baseline, recent studies have demonstrated that kappa coefficient varies with geophysical conditions. Angling et al. (2018) first proposed an empirical model, $\kappa(a) = f(F10.7, \chi, a)$, approximating it as a linear function of solar flux index (F10.7), solar zenith angle (χ), and impact parameter (a). Park et al. (2025) subsequently extended this approach as $\kappa(a) = f(F10.7, \chi, a, \lambda_{mag}, t_{LT})$ by incorporating geomagnetic latitude (λ_{mag}) and local time (t_{LT}) into the linear approximation and performed separate linear regressions for four conditions (day/night × high/low solar activity) to better capture complex spatiotemporal variations.

Despite these advancements, both the analytical and empirical kappa models are fundamentally constrained by the assumption of spherical symmetry. Consequently, complex ionospheric asymmetries driven by strong horizontal gradients introduce systematic biases in individual profiles, causing α_{corr} to deviate from the true α_{NA}. This deviation degrades the accuracy of long-term climatological trends, particularly during periods of high solar activity.

2.2 Dataset Construction

To train and validate the deep learning model, we constructed a "ground truth" RIE dataset by performing 3D ray-tracing simulations using the COSMIC-2 constellation geometry, specifically limiting the analysis to GPS occultations. The simulation environment was configured to reflect the ionospheric conditions of 2024 (covering day of year 1 to 226), corresponding to the maximum phase of Solar Cycle 25. We utilized the NeQuick ionospheric electron density model, driven by daily observed F10.7 solar flux indices (ranging approximately from 120 to over 200 sfu) to ensure realistic representations of the ionosphere. From the daily occultation events, we randomly sampled a subset of profiles to ensure diverse geometrical and temporal coverage while maintaining computational efficiency.

The dataset processing pipeline involved the following key steps:

1. 3D ionospheric ray-tracing (ground truth data generation): To rigorously isolate the ionospheric effects, we performed ray-tracing simulations using the Radio Occultation Simulator for Atmospheric Profiling (ROSAP) algorithm (Høeg et al., 1995).

GPS and LEO satellite positions were obtained from CDAAC Precise Orbit Determination data. The simulation environment was constructed using only the ionospheric refractive index derived from 3D electron density values provided by NeQuick, effectively excluding the neutral atmosphere. Since there are no neutral atmospheric contributions to the signal bending in this simulation environment, the theoretical neutral atmospheric bending angle (α_{NA}) is zero. Consequently, substituting the simulated L1 and L2 bending angles into the standard linear combination equation (Eq. 1) yields the RIE (α_{RIE}) as the remaining results.

2. Coordinate systems and metadata: Tangential point coordinates of occultation cases were used to calculate auxiliary geophysical parameters. Solar zenith angles (SZA) were computed using the NOAA solar position algorithm (Reda and Andreas, 2004), and geomagnetic latitudes were derived using the Adaptive AACGM-v2 (Altitude Adjusted Corrected Geomagnetic Coordinates Model) (Shepherd, 2014) to accurately approximate low-latitude values.

3. Data filtering: The analysis focuses on the impact height range of 40-80 km covering the upper stratosphere and mesosphere. In this region, the magnitude of the neutral atmospheric bending angle is relatively small, making the relative impact of RIE a dominant error source. Profiles containing non-finite values or physically unrealistic RIE magnitudes (specifically the extreme 0.5% of the data distribution) caused by simulation artifacts were excluded.

The final dataset comprises a total of 3,865 occultation events. The simulations cover an impact height with a median vertical resolution of approximately 958 m (mean: 932 m). The ground truth RIE values in the dataset range from approximately -2.8×10^{-1} μrad to $+2.4 \times 10^{-1}$ μrad, capturing both the typical negative biases and the anomalous positive biases associated with asymmetrical conditions.

2.3 Feature Engineering and Input Representation

A critical limitation of the existing theoretical model, kappa correction, is its reliance on the assumption of spherical symmetry. To empower the deep learning model to capture systematic biases arising from spherical asymmetry, we introduce two additional geometric features: ray-path azimuth and bending angle slope.

First, the ray-path azimuth is derived to account for the directional dependency of the signal path relative to ionospheric gradients. Using the Earth-centered inertial coordinates of the GPS (r_{GPS}) and LEO (r_{LEO}) satellites obtained from the ephemeris data, we computed the line-of-sight (LOS) vector and transformed it into the local East-North-Up frame at the occultation point. To preserve angular continuity during training, the azimuth angle ϕ is converted into cyclical components:

$$X_{azimuth} = [\sin(\phi), \cos(\phi)] \tag{4}$$

where ϕ denotes the azimuth angle of the ray path projected onto the local horizontal plane, measured clockwise from the North.

Second, we incorporate the bending angle slope (β) as a proxy for the event-specific ionospheric intensity. The algorithm implements a linear regression of the squared bending angle difference $(\alpha_1 - \alpha_2)^2$ with respect to altitude for each event. The slope derived from this regression is then designated as a representative value characterizing the overall ionospheric state of that specific occultation case.

Final input dimension is $D = 17$, consisting of the following features for each altitude step (to preserve continuity in angular variables, we applied sine/cosine encoding):

- Physics baseline features (4): The standard kappa coefficient (Eq. 3), the theoretical RIE estimate based on kappa correction, the bending angle difference, and the squared bending angle difference.
- Signal pattern features (2): Event-specific bending angle difference slope (β), squared bending angle difference slope (β^2).
- Satellite geometry features (4): Ray-path azimuth ($\sin(azimuth), \cos(azimuth)$), LEO perpendicular velocity (V_{mean}), and normalized LEO altitude.
- Geophysical context (7): Geomagnetic latitude (λ_{mag}), solar flux (F10.7), local time ($\sin(t), \cos(t)$), signal altitude, and SZA ($\sin(SZA)$, $\cos(SZA)$).

2.4 Proposed Model Architecture: PhyG-TRF

We propose PhyG-TRF (Physics-Guided Transformer), as illustrated in Figure 1, a specialized altitudinal sequence-to-sequence learning architecture designed to mitigate RIE by capturing both the altitudinal dependencies and the global environmental contexts of the occultation. The model employs a residual sequence learning strategy, where the "sequence" corresponds to the vertical profile of a single occultation event. The input matrix $X \in \mathbb{R}^{L\times D}$ consists of L altitude steps with $D = 17$ features. Since the number of data points varies per event (e.g., due to different satellite geometry changes), we utilize a padding and boolean masking mechanism to ensure that gradient updates are calculated solely based on valid physical measurement points.

Adopting the Hybrid-Physics-Data Residual (HPD-Res) modeling strategy from the PGNN framework (Daw et al., 2021), we augment the input matrix X with the theoretical RIE estimate derived from the kappa correction model. By explicitly providing the physics-based estimation as a feature, the model is conditioned to learn the residual correction from the physics baseline to the true RIE. The raw input sequence is first processed by a linear projection layer to map the 17-dimensional features into a latent model dimension of $d_{model} = 128$. To preserve the order of the altitude sequence, we utilize sinusoidal positional encodings. Unlike standard Transformers that use additive encoding, we employ a concatenation-based strategy to preserve the continuous signal features. The sinusoidal positional encodings are concatenated with the projected embeddings, followed by a second linear transformation to restore the feature dimension to d_{model}. This design ensures that the relative vertical ordering is encoded without distorting the magnitude of the physical features.

The core backbone of PhyG-TRF consists of $N = 3$ stacked Transformer encoder layers. Each layer employs a Multi-Head Self-Attention (MHSA) mechanism with $h = 4$ heads (Vaswani et al., 2017) to construct a global receptive field across the altitude profile. Unlike Recurrent Neural Networks (RNNs) that process data sequentially, MHSA allows the prediction at a specific altitude to be informed by the ionospheric state at all other altitudes simultaneously within the occultation profile. Mathematically, for a given altitude query Q, the mechanism computes a weighted sum of values V based on the similarity between the query and keys K derived from the entire profile:

$$\mathrm{Attention}(\mathrm{Q}, \mathrm{K}, \mathrm{V}) = \mathrm{softmax}\left(\frac{\mathrm{QK}^{\mathrm{T}}}{\sqrt{\mathrm{d_k}}} + M\right)\mathrm{V} \tag{5}$$

where d_k is the dimension of the key vector, and M is the attention mask matrix that sets the attention scores of padded tokens to $-\infty$. This ensures that the learned correlations represent genuine vertical dependencies—such as the propagation of bending angle perturbations across ionospheric layers—rather than artifacts from zero-padding.

To enhance training stability, we adopt the Pre-Layer Normalization (Pre-LN) architecture. In this configuration, LN is applied before the MHSA and Feed-Forward Network (FFN) blocks. The output of the l-th layer is formulated as:

$$x_l' = x_l + \mathrm{MHSA}(LN(x_l)) \tag{6}$$

$$x_{l+1} = x_l' + \mathrm{FFN}(LN(x_l')) \tag{7}$$

The FFN consists of two dense layers with an inner dimension of $d_{ff} = 256$. We employ the Rectified Linear Unit (ReLU) activation function, defined as $ReLU(x) = max(0, x)$. The FFN operation is expressed as:

$$\mathrm{FFN}(x) = \max(0, x \cdot W_1 + b_1) \cdot W_2 + b_2 \tag{8}$$

where W_1, b_1 and W_2, b_2 represent the weights and biases of the two linear transformations, respectively. A dropout rate of 0.1 is applied to the output of each sub-layer.

A distinct feature of our architecture is the Masked Global Context Layer (MGCL). While the encoder captures local vertical variations, the overall RIE magnitude is strongly governed by static environmental conditions. To explicitly model this, we extract a global context vector c_{global} via a masked average pooling operation directly over the raw input matrix X. This ensures that the model captures the physical features (e.g., SZA, solar flux) independent of the latent transformations. This vector is processed by a projection MLP via the following equation:

$$c_{global} = ReLU(W_p \cdot MaskedAvgPool(X) + b_p) \tag{9}$$

where W_p and b_p are the learnable parameters of the projection layer. This context vector is replicated L times to match the sequence length.

The final prediction is performed by fusing the local and global representations. The sequence output from the encoder, the replicated global context vector, and the physics baseline are concatenated and passed through a regression head to compute a residual correction $\Delta(a)$. The head consists of two dense layers (64 and 32 units) with ReLU activation and dropout. The final output is produced by a VARL, which combines the physics baseline and the learned residual through a learned scalar gate $g(a) \in [0, 2]$:

$$\alpha_{RIE}(a) = g(a) \cdot \alpha_{kappa}(a) + \Delta(a) \tag{10}$$

where $g(a) = sigmoid(\cdot) \times 2.0$ is computed from the concatenation of the physics baseline and the Transformer sequence output, enabling the model to adaptively weight the physics baseline against the data-driven residual correction for each altitude step. Unlike the regression head that directly predicts RIE, the VARL formulation anchors the prediction to the physics model while allowing the network to learn complex residual mapping.

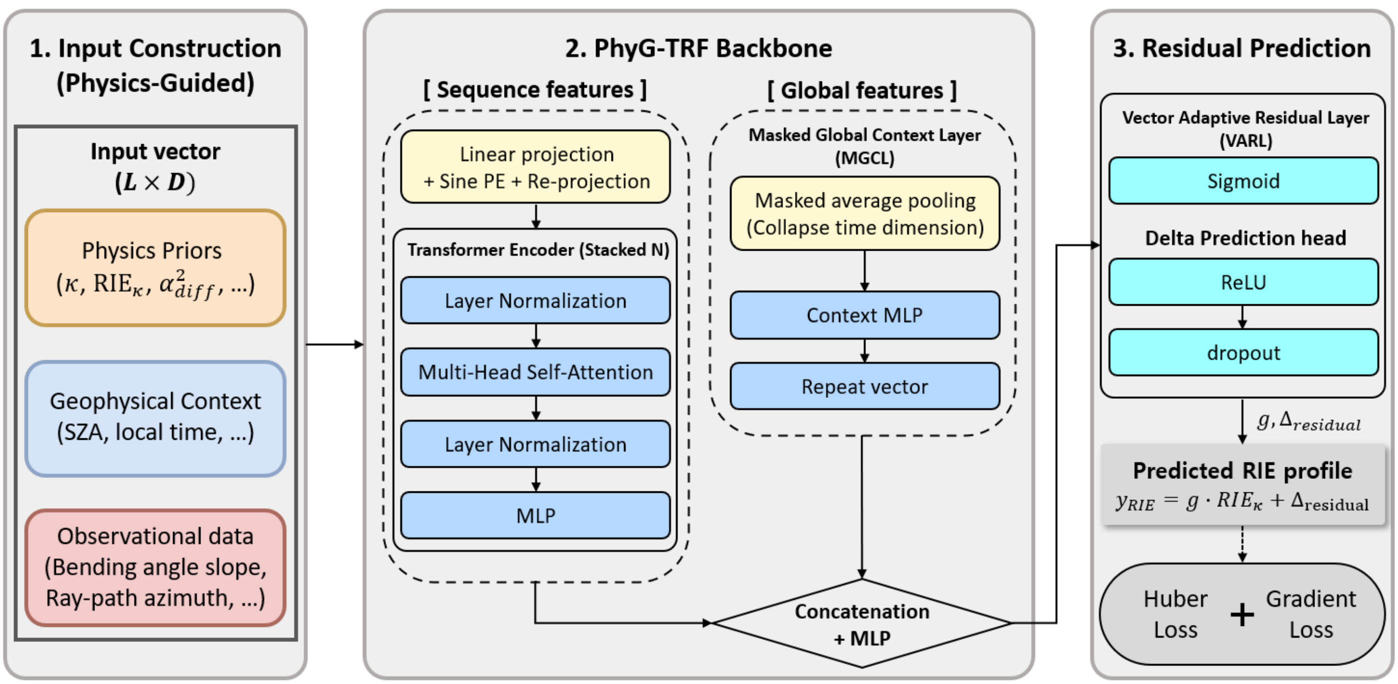

FIGURE 1 PhyG-TRF Architecture Diagram showing inputs from the data pipeline feeding into the Transformer blocks

2.5 Loss Function and Case-Level Adaptive Weighting

Training employs a composite loss combining Huber loss ($\delta = 0.5$) on the predicted RIE and Mean Squared Error (MSE) on the predicted RIE gradient with respect to altitude, with a gradient loss coefficient $\lambda = 0.5$. Padding tokens are excluded from all loss computations:

$$L = L_{Huber}(\hat{y}, y) + \lambda \cdot MSE(\frac{\partial \hat{y}}{\partial h}, \frac{\partial y}{\partial h}) \tag{11}$$

To address the underrepresentation of asymmetric (positive-trending) RIE cases in the training data, we apply a case-level adaptive weighting scheme based on the positive gradient ratio, defined as the fraction of altitude steps where the RIE slope is positive. Case weights are computed from the empirical CDF of positive gradient ratio among positive-trending events, scaled to a maximum weight of $w_{max} = 5.0$. Events with positive gradient ratio = 0 (symmetric, negative-trending RIE) receive a weight of 1.0.

3 EXPERIMENTAL SETUP

To evaluate the efficacy of the proposed PhyG-TRF framework, we conducted comparative experiments against established physics-based methods and other machine learning architectures. This section details the baseline models, the training and validation strategies employed to prevent data leakage, and the specific learning paradigms investigated.

3.1 Baselines

We compare the performance of PhyG-TRF against four baseline categories, allowing us to isolate the contributions of physics knowledge, empirical kappa refinement, non-linear mapping, and sequence learning capabilities.

- Physics Baseline (Healy & Culverwell, 2015): As the standard, we utilize the analytical kappa correction model described in Section 2.1 with kappa coefficient term in Equation (3). This model serves as the reference for evaluating how well data-driven approaches can improve upon a theoretical-driven method that assumes spherical symmetry. The corrected bending angle is calculated using Equation (2), and the corresponding RIE is derived analytically.

- Empirical Kappa Correction: To assess the headroom available to physics-based approaches, we also include two empirical kappa models. The Angling model (Angling et al., 2018) approximates the kappa coefficient as a linear function of solar flux (F10.7), solar zenith angle, and impact parameter. The Park et al. (2025) model extends this with a four-condition regression (day/night × high/low F10.7) that additionally incorporates geomagnetic latitude and local time, capturing more complex spatiotemporal variations in the kappa coefficient.

- XGBoost (Tabular baseline): To investigate the performance of a strong non-sequential learner, we implemented an XGBoost (Extreme Gradient Boosting) regressor. This model treats each altitude point independently, ignoring the vertical ordering of the profile. It was configured with 1,000 estimators, a maximum depth of 9, and a learning rate of 0.05. This baseline helps quantify the gain achieved purely by complex non-linear mapping without explicit sequence modeling.

- Bi-directional GRU (Sequence baseline): To evaluate the advantage of the Transformer's self-attention mechanism over traditional RNN architectures, we implemented a Bi-directional Gated Recurrent Unit (Bi-GRU) network. The model consists of 2 stacked Bi-GRU layers with 64 units each. While the Bi-GRU can capture vertical dependencies, it processes the altitude profile sequentially (step-by-step), unlike the Transformer which processes the global context in parallel.

Both XGBoost and Bi-GRU are trained under both HPD-Res (residual) and direct learning paradigms. The HPD-Res variant of each model is used for the main performance comparison in Section 4.1; a dedicated ablation comparing residual versus direct learning across all three architectures is presented in Section 4.2.

TABLE 1
Summary of models and their input contexts

Model Type	Architecture Details	Input Context
Kappa correction (Physics baseline)	Analytical parametric model	Point-wise
Angling et al. (2018) (Empirical)	Linear regression κ (F10.7, SZA, altitude)	Point-wise

Park et al. (2025) (Empirical)	4-condition regression κ (F10.7, SZA, altitude, local time, geomagnetic latitude)	Point-wise
XGBoost (ML baseline)	Gradient boosting trees (n=1000, depth=9)	Point-wise (Tabular)
Bi-GRU (Deep learning)	Bidirectional GRU (2 layers, 64 units each)	Local sequence (Recurrent)
PhyG-TRF (Deep learning)	Transformer encoder (N=3, h=4, dmodel=128) + MGCL + VARL	Global sequence (Attention)

3.2 Data Partitioning Strategy

A critical challenge in training GNSS-RO machine learning models is the high correlation of data points within a single occultation event. Standard random splitting strategies would scatter points from the same vertical profile across training and testing sets, leading to severe data leakage and over-optimistic performance estimates. To prevent this, we employed a group-wise splitting strategy based on unique events, ensuring that all altitude points belonging to a specific occultation event are assigned exclusively to a single subset.

Consequently, the dataset was partitioned using a hierarchical approach to achieve an approximate 64:16:20 ratio, ensuring robust evaluation on unseen events. The training set comprises 2,473 events (110,316 points), representing roughly 64% of the data, and was used for model fitting. The validation set includes 619 events (27,901 points), accounting for approximately 16%, which was utilized for hyperparameter tuning and monitoring early stopping criteria. Finally, the test set consists of 773 events (35,026 points), making up the remaining 20%, and was strictly held out for the final performance evaluation. We explicitly verified that there is zero overlap of event IDs between these subsets to guarantee the integrity of the evaluation.

3.3 Training Details

Deep learning models were implemented using the TensorFlow framework, while XGBoost was trained using the official XGBoost library. To determine the optimal model configuration for the proposed PhyG-TRF, we conducted Bayesian hyperparameter optimization using the KerasTuner library. The training process for both PhyG-TRF and Bi-GRU utilized the Adam optimizer with an initial learning rate of 5×10^{-4}. To ensure training stability—particularly for the Transformer and Recurrent architectures—we applied gradient clipping with a norm value of 1.0, which prevents gradient explosion during backpropagation.

The batch size was set to 32, and the maximum number of training epochs was set to 120. To promote convergence and prevent overfitting, we employed dynamic callback strategies: Early Stopping halted training if the validation loss did not improve for 15 consecutive epochs, restoring the best model weights. Furthermore, to facilitate fine-grained convergence as the model approached a minimum, we utilized an adaptive learning rate scheduler (implemented via the ReduceLROnPlateau callback) to reduce the learning rate by a factor of 0.5 if the validation loss plateaued for 5 epochs.

4 RESULTS AND DISCUSSION

In this section, we present a comprehensive evaluation of the proposed PhyG-TRF framework. We first provide a quantitative comparison against the physics-based and empirical baselines and other machine learning architectures. We then validate the HPD-Res residual learning strategy, analyze the distinct advantages of the altitudinal sequence learning approach over point-wise learning through jitter analysis, and examine case studies and error distributions. Finally, we assess performance stratified by ionospheric asymmetry strength.

4.1 Overall Performance Evaluation

The overall performance of all models was evaluated on the held-out test set (34,740 data points from 773 events). Table 2 summarizes the statistical metrics, including Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Correlation (R), the Coefficient of Determination (R^2), RMSE Skill relative to the kappa correction, and mean bias.

TABLE 2
Comparison of RIE correction performance across models

Model Type	RMSE (10^{-2}μrad)	MAE (10^{-2}μrad)	R^2 Score	Skill (%)	Correlation	Bias (10^{-2}μrad)
Kappa correction (Physics baseline)	2.892	1.709	0.254	0.0	0.838	−1.343
Angling et al. (2018)	3.189	1.974	0.093	−10.2	0.833	−1.759
Park et al. (2025)	2.523	1.515	0.432	+12.7	0.820	−0.985
XGBoost (ML baseline)	1.513	0.899	0.796	+47.7	0.894	+0.149
Bi-GRU (Deep learning)	1.496	0.934	0.800	+48.3	0.895	+0.096
PhyG-TRF (Deep learning)	1.387	0.843	0.828	+52.1	0.912	+0.021

As shown in Table 2, the baseline kappa correction exhibits an RMSE of $2.892 \times 10^{-2}\ \mu rad$ and a low R^2 of 0.254. This limited performance confirms that the spherically symmetric assumption of the kappa model is insufficient to capture the complex RIE variability present in the 2024 solar maximum simulation data. Under the high solar activity and asymmetric ionospheric conditions of this study, the Angling et al. (2018) model performs slightly worse than the physics baseline (RMSE: $3.189 \times 10^{-2}\ \mu rad$, skill: -10.2%), suggesting that its linear coefficients, derived under more moderate conditions, do not generalize well to the strong horizontal gradient regime. The Park et al. (2025) model, despite incorporating geomagnetic latitude and local time, achieves only a modest improvement (skill: $+12.7\%$), reflecting the inherent limitations of linear kappa models under asymmetric ionospheric conditions. In contrast, all data-driven machine learning models achieve substantially higher performance than the physics baselines.

Among the ML models, the two altitudinal sequence learning models—PhyG-TRF and Bi-GRU—achieve higher R^2 scores (0.828 and 0.800, respectively) than the point-wise XGBoost (0.796), confirming that capturing vertical dependencies across the altitude profile provides a meaningful advantage. PhyG-TRF achieves the lowest RMSE of $1.387 \times 10^{-2}\ \mu rad$, corresponding to a 52.1% reduction compared to the kappa correction. Furthermore, the R^2 score improves to 0.828, indicating that the Transformer architecture effectively explains the variance in the RIE ground truth. Notably, PhyG-TRF outperforms the Bi-GRU, demonstrating that the global self-attention mechanism is more effective than sequential recurrent processing in capturing the vertical dependencies of ionospheric errors.

4.2 Validation of Residual Learning Strategy

A key design decision underlying PhyG-TRF is the adoption of the HPD-Res learning strategy: the physics-based kappa correction is provided as an explicit input feature, and the model is trained to predict the residual error relative to this baseline. To validate this choice, we compare residual and direct learning variants across all three machine learning architectures used in this study: PhyG-TRF, Bi-GRU, and XGBoost.

In the direct learning paradigm, the model predicts the absolute RIE value from the 17-feature input without any physics-based anchor. In the residual learning (HPD-Res) paradigm, the kappa correction estimate is included as an explicit feature and the model targets the residual between the true RIE and the physics baseline, with the final prediction reconstructed by adding the kappa correction back.

Figure 2 illustrates the RMSE comparison between residual and direct learning variants for each architecture. Residual learning consistently outperforms direct learning across all three architectures—PhyG-TRF, Bi-GRU, and XGBoost—with the kappa baseline shown as reference. For PhyG-TRF, the residual variant achieves an RMSE of $1.387 \times 10^{-2}\ \mu rad$ (skill: 52.1%) versus $1.419 \times 10^{-2}\ \mu rad$ (skill: 50.9%) for the direct variant, a 2.2% relative improvement. For Bi-GRU, the improvement is more substantial: $1.496 \times 10^{-2}\ \mu rad$ (48.3%) versus $1.645 \times 10^{-2}\ \mu rad$ (43.1%), a 9.0% relative improvement. For XGBoost, the residual variant achieves $1.513 \times 10^{-2}\ \mu rad$ (47.7%) versus $1.543 \times 10^{-2}\ \mu rad$ (46.7%), a 1.9% relative improvement. This consistent pattern across architecturally distinct models rules out the possibility that the improvement is an artifact of any single model's inductive bias; instead, it reflects a fundamental advantage of anchoring data-driven learning to an established physics baseline.

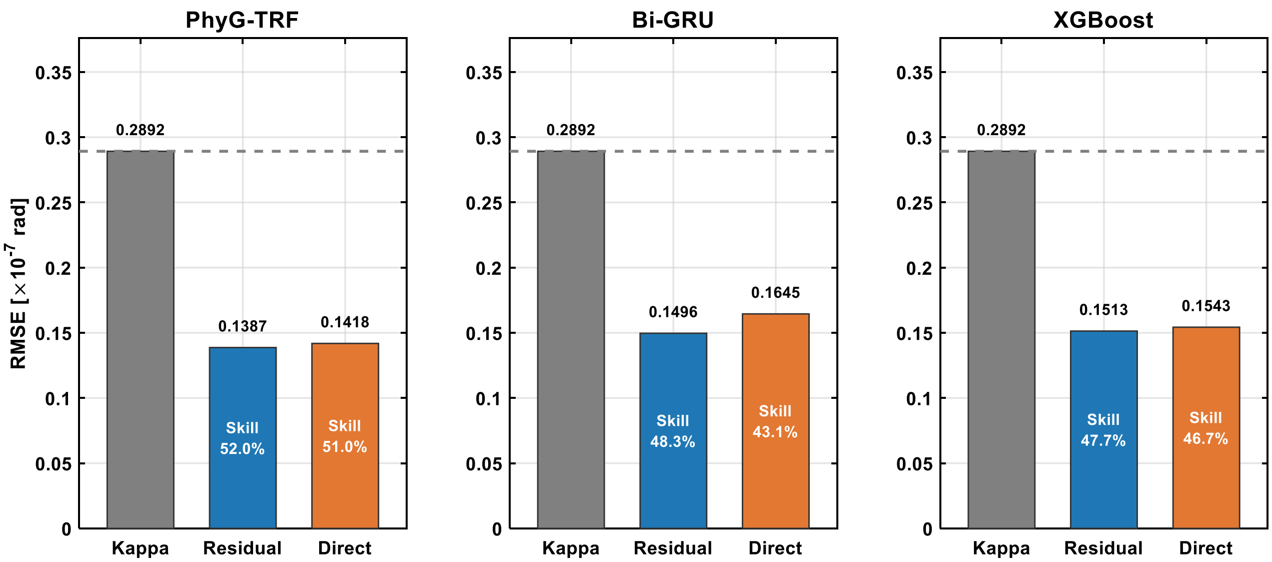

FIGURE 2 RMSE comparison between residual (HPD-Res) and direct learning variants for PhyG-TRF, Bi-GRU, and XGBoost

4.3 Case Study: Negative vs. Positive RIE Estimation

Figure 3 presents representative case studies for a negative-trending and a positive-trending occultation event, comparing the kappa correction baseline and PhyG-TRF predictions against the simulated ground truth RIE. In the negative-trending RIE case, both the kappa correction and PhyG-TRF track the ground truth reasonably well, as the RIE exhibits a monotonically decreasing profile with altitude that is broadly consistent with the kappa correction's design assumption. In the positive-trending RIE case, however, the kappa correction produces a negative prediction that is opposite in sign to the ground truth—actively worsening the retrieval—while PhyG-TRF correctly captures the positive and increasing RIE profile. This positive-trending behavior, driven by strong horizontal electron density gradients, represents the primary failure mode of physics-only models and constitutes the core motivation for the proposed altitudinal sequence learning approach.

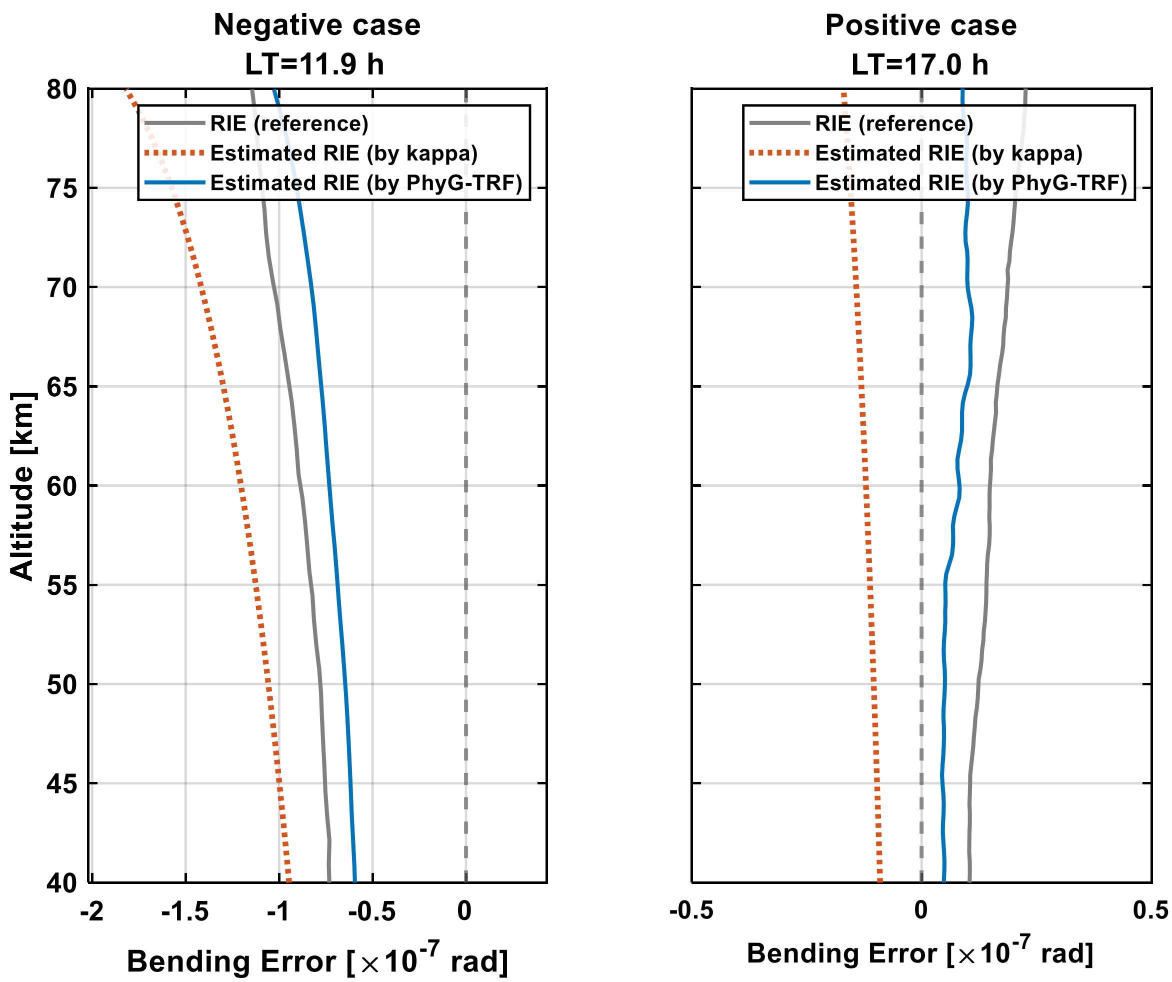

FIGURE 3 Case study vertical RIE profiles for a negative-trending case (left) and a positive-trending case (right)

4.4 Prediction Accuracy and Correlation Analysis

The overall predictive capabilities of the models are qualitatively illustrated in Figure 4. The scatter plots reveal that while the kappa correction exhibits broad dispersion with a correlation coefficient of $R = 0.838$ and $R^2 = 0.254$, the data-driven models demonstrate substantially superior alignment with the ground truth. PhyG-TRF achieves the highest correlation of $R = 0.912$ and $R^2 = 0.828$, showing the tightest clustering along the identity line—particularly in the positive-RIE regime where the kappa correction systematically produces predictions of the wrong sign. The performance ordering, PhyG-TRF > Bi-GRU > XGBoost ($R^2 = 0.828 > 0.800 > 0.796$), reflects the hierarchy of the altitudinal sequence learning approach: global self-attention captures the full vertical context simultaneously, recurrent processing captures local sequential dependencies, and point-wise processing ignores vertical ordering entirely.

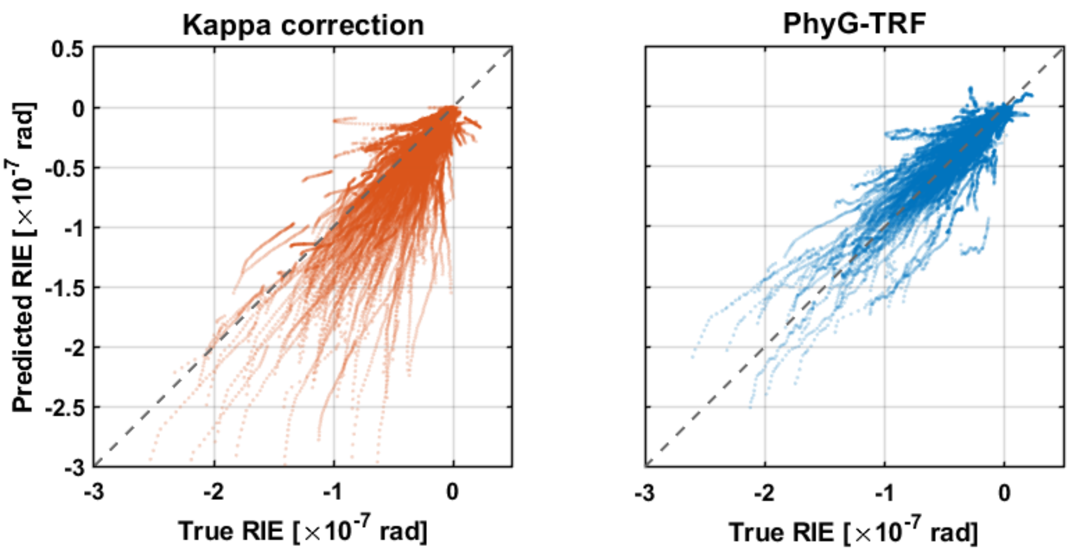

FIGURE 4 Scatter plots of predicted RIE vs. true RIE for the full test set. Left: kappa correction ($R^2 = 0.254$). Right: PhyG-TRF ($R^2 = 0.828$)

4.5 Analysis of Sequence-wise vs. Point-wise Learning

While the overall RMSE metrics demonstrate the predictive power of machine learning models over the physics baseline, a critical distinction arises from their structural learning paradigms: point-wise (XGBoost) versus sequence-wise (PhyG-TRF, Bi-GRU) estimation. Since RIE profiles represent continuous physical phenomena, structural continuity—specifically the absence of non-physical high-frequency noise—is as crucial as the absolute magnitude of the error. Additionally, the altitude region targeted for RIE correction is inherently susceptible to measurement noise due to the lower magnitude of neutral atmospheric bending angle. Therefore, employing a noise-susceptible model for correction can affect the final retrieval accuracy, potentially yielding atmospheric profiles with artifacts that are inferior to the uncorrected baseline, despite statistical improvements in RIE RMSE.

To quantify the physical consistency of the predicted profiles, we introduced a Jitter Metric (J). Since ionospheric profiles naturally exhibit smooth variations with altitude, high-frequency fluctuations in the prediction often indicate non-physical artifacts. We isolated these fluctuations by decomposing the predicted profile $\hat{y}$ into a smooth physical trend y_{trend} and a high-frequency residual noise component using a Savitzky-Golay (SG) filter as in Equation (12).

$$y_{trend} = SG(\hat{y}, \text{window} = 11, \text{order} = 3) \tag{12}$$

The Jitter Metric is defined as the RMS magnitude of the high-frequency residual.

$$J(\hat{y}) = \sqrt{\frac{1}{L}\sum_{i=1}^{L}\left(\hat{y}_i - y_{trend,i}\right)^2} \tag{13}$$

where L is the number of altitude steps in the profile.

Table 3 and Figure 5 illustrate a representative worst-case scenario where the difference between learning paradigms is most pronounced. The XGBoost (Point-wise) model exhibits sawtooth-like fluctuations, failing to maintain vertical continuity. This occurs because the point-wise learner treats each altitude step h_i as an independent instance, ignoring the strong correlation with adjacent altitudes $h_{i\pm1}$. Consequently, it overfits local feature variations, resulting in discontinuous artifacts. In contrast, the PhyG-TRF (Sequence-wise) model produces a smooth, continuous profile that closely follows the ground truth. By leveraging the self-attention mechanism, the sequence-wise model aggregates context from the entire vertical profile, effectively filtering out local inconsistencies.

TABLE 3
Comparison of structural jitter (high-frequency noise) across different RIE correction models

Model	Jitter Metric (J) ($\times 10^{-10}$)	Ratio to True RIE
True RIE	2.79	1.00
Kappa correction (Physics baseline)	0.12	0.04
XGBoost (Point-wise)	6.61	2.37
Bi-GRU (Sequence-wise)	2.08	0.74
PhyG-TRF (Sequence-wise)	2.14	0.77

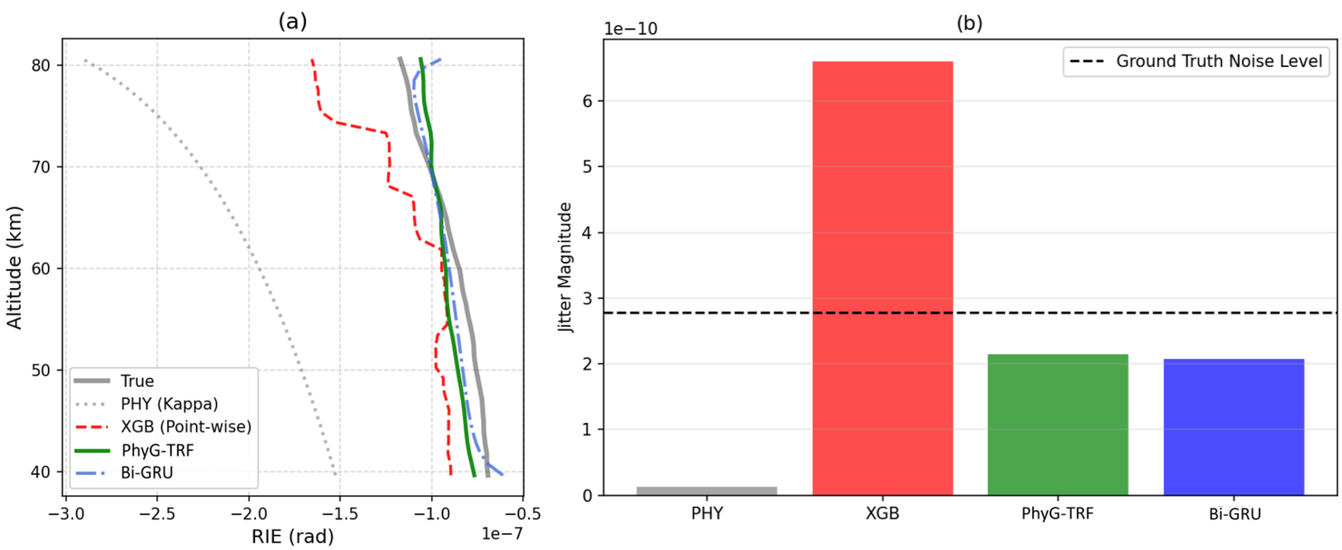

FIGURE 5 (a) Vertical RIE profiles for an example case, illustrating the structural instability of the point-wise model (XGB, red dashed) compared to the smooth reconstruction of the sequence-wise model (PhyG-TRF, green solid; Bi-GRU, blue dashed). The gray solid line represents the ground truth. (b) Comparison of average jitter magnitude across all test cases. The horizontal black dashed line indicates the intrinsic noise level of the ground truth profiles.

Statistical results highlight a fundamental limitation of the point-wise approach. The XGBoost model exhibits an average jitter of 6.61×10^{-10}, which is 2.37 times higher than the intrinsic noise level of the ground truth. This suggests that the tabular model amplifies input noise, creating artificial discontinuities. Conversely, the sequence-wise models (PhyG-TRF and Bi-GRU) achieve jitter ratios of 0.77 and 0.74, respectively. A ratio below 1.00 indicates that these models not only avoid generating artificial noise but also demonstrate a restorative capability, smoothing out some of the thermal noise present in the ground truth labels while preserving the physical gradients. However, an excessively low ratio like that of the physics model is not necessarily ideal, as it may indicate over-smoothing where genuine ionospheric variations are suppressed along with the noise. Thus, the observed ratios represent an optimal balance between noise reduction and signal preservation.

4.6 Error Distribution

The overall predictive capabilities are further substantiated by the error distribution histograms in Figure 6, where the gray region indicates the RIE distribution without any correction, the orange distribution represents the remaining error after the kappa correction, and the blue distribution represents the remaining error after the PhyG-TRF correction. On the full test set, the kappa correction error is negatively skewed and broadly distributed (mean bias: $-1.343 \times 10^{-2}\ \mu rad$). In contrast, the PhyG-TRF prediction error is sharply concentrated near zero, achieving a 52.1% RMSE reduction and a 98.5% mean bias reduction (bias: $+0.021 \times 10^{-2}$ μrad) relative to the kappa correction (physics baseline)—confirming its superior capability in minimizing both systematic bias and random fluctuations. On the positive-trending subset, the kappa correction error is even more strongly negatively skewed—as the model predicts the wrong sign of the RIE—while PhyG-TRF reduces the RMSE by 70.0% and the mean bias by 80.9% (from -1.167×10^{-2} μrad to -0.223×10^{-2} μrad).

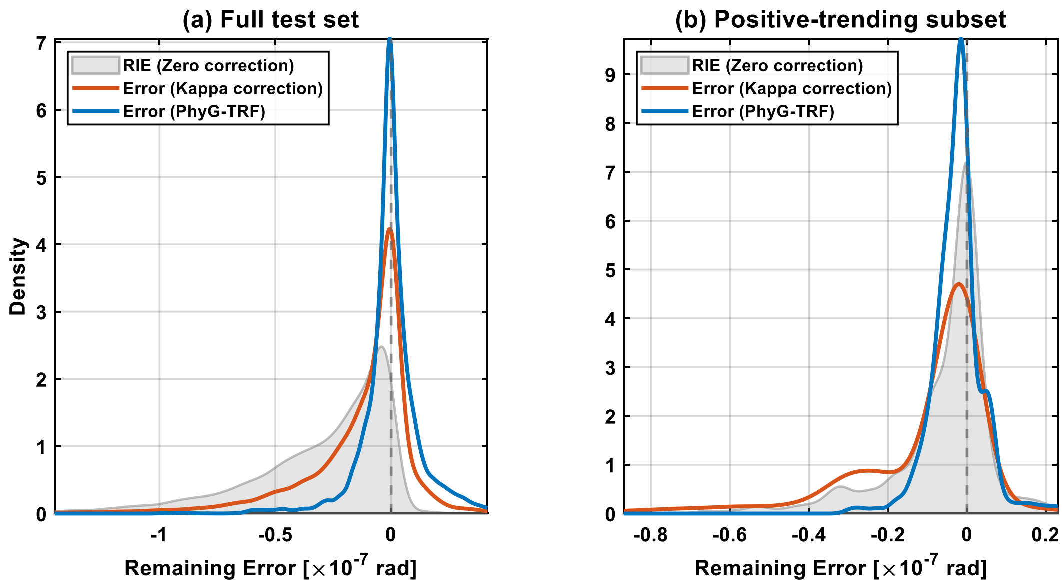

FIGURE 6 Error distribution plots (a) Full test set (b) Positive-trending RIE subset

4.7 Performance under Varying Asymmetry Strength

Table 4 presents the RMSE skill of PhyG-TRF over the kappa correction as a function of the positive gradient ratio, which serves as a proxy for ionospheric asymmetry strength.

TABLE 4
PhyG-TRF RMSE skill (%) and R² over the kappa correction, stratified by positive gradient ratio band

Positive Ratio Band	Band Definition	N (timesteps)	RMSE Skill (%)	R^2
Band 0	r = 0 (symmetric)	19,124	+49.7	0.806
Band 1	$0 < r \leq Q50$	7,670	+56.4	0.705
Band 2	$Q50 < r \leq Q75$	5,233	+52.8	0.607
Band 3	$Q75 < r \leq Q90$	1,353	+67.5	0.755
Band 4	$r > Q90$ (most asymmetric)	1,360	+71.3	0.605

PhyG-TRF consistently outperforms the kappa correction across all five asymmetry bands. The RMSE skill improves from +49.7% for symmetric cases (Band 0) to +71.3% for the most strongly asymmetric cases (Band 4), confirming that the CDF-based case-level adaptive weighting scheme effectively prioritizes the underrepresented asymmetric cases during training. The non-monotonic pattern between Bands 1–3 reflects the varying complexity of partially asymmetric ionospheric structures in the intermediate regime.

5 CONCLUSION

This study proposed PhyG-TRF (Physics-Guided Transformer), a physics-guided deep learning framework designed to mitigate residual ionospheric errors in GNSS-RO bending angles. Unlike traditional analytical models constrained by the assumption of spherical symmetry, our approach integrates the physics baseline of the kappa correction with a data-driven altitudinal sequence learning architecture. By adopting the HPD-Res strategy and incorporating global environmental context—such as solar zenith angle, geomagnetic latitude, and ray-path azimuth—through a Masked Global Context Layer (MGCL) and a Multi-Head Self-Attention (MHSA) mechanism, the proposed model effectively captures the complex nonlinear biases arising from ionospheric asymmetry. A Vector Adaptive Residual Layer (VARL) enables adaptive blending of the physics baseline and the learned residual, while a composite gradient-aware loss function and CDF-based case-level adaptive weighting further reinforce training robustness for asymmetric cases.

Through a comprehensive evaluation using 3D ray-tracing simulations corresponding to the solar maximum of 2024, we demonstrated that PhyG-TRF outperforms both the physics baseline and other machine learning correction methods. Quantitative analysis revealed that PhyG-TRF achieves a 52.1% RMSE reduction over the kappa baseline on the full test set (R^2 improving from 0.254 to 0.828). On the positive-trending RIE subset—where the kappa correction actively degrades retrievals with an R^2 of -2.52—PhyG-TRF achieves an RMSE reduction of 70.0%, an R^2 of 0.684, and an 80.9% reduction in mean bias, confirming that the physics-guided strategy successfully resolves systematic biases that analytical and empirical kappa models fail to capture, particularly under conditions of strong horizontal electron density gradients.

Furthermore, our comparative analysis of learning paradigms highlighted the advantage of sequence-wise learning over point-wise approaches. While the tabular baseline (XGBoost) exhibited high-frequency artifacts, amplifying intrinsic noise by a factor of 2.37, PhyG-TRF maintained physical consistency with a jitter ratio of 0.77. This demonstrates that the altitudinal sequence learning approach not only corrects the RIE magnitude but also preserves the structural smoothness of the bending angle profiles, which is essential for accurate atmospheric retrievals. Additionally, the systematic comparison of residual versus direct learning variants across all three architectures confirmed that the HPD-Res physics conditioning strategy consistently improves performance regardless of the underlying model family, providing empirical validation for the physics-guided design principle.

In our future work, we aim to validate PhyG-TRF against operational weather model outputs (e.g., ECMWF) using real GNSS-RO measurements, which will provide a critical assessment of the model's generalization beyond simulation-based evaluation. Furthermore, we plan to extend the training dataset to cover a broader range of solar cycle phases and geomagnetic conditions, progressively enhancing the model's generalization capability and correction accuracy. By leveraging these expanded ground truth data, we seek to further minimize the difference between predicted and actual RIEs, ultimately contributing to more reliable climate monitoring and numerical weather prediction systems.

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